

Econ 771 hw 1

1. if we know that D is independent with ε conditional on X ,

$$\text{then we have } f_{\varepsilon, D|X}(e, d|x) = f_{\varepsilon|x}(e|x) \cdot f_{D|x}(d|x)$$

$$\text{by the definition, we have } f_{\varepsilon|D, X}(e|d, x) = \frac{f_{\varepsilon, D, X}(e, d, x)}{f_{D, X}(d, x)}$$

$$\begin{aligned} f_{\varepsilon, D, X}(e, d, x) &= f_{\varepsilon, D|x}(e, d|x) \cdot f_X(x) = f_{\varepsilon|x}(e|x) f_{D|x}(d|x) \cdot f_X(x) \\ f_{D, X}(d, x) &= f_{D|x}(d|x) \cdot f_X(x) \end{aligned} \quad \left. \begin{array}{l} \text{as } X \text{ (controls)} \\ \text{is indep. from } D \\ \text{(treatments) and } \varepsilon \\ \text{(unobservables)} \end{array} \right\}$$

$$\text{Hence, } f_{\varepsilon|D, X}(e|d, x) = \frac{f_{\varepsilon|x}(e|x) \cdot f_{D|x}(d|x) \cdot f_X(x)}{f_{D|x}(d|x) \cdot f_X(x)} = f_{\varepsilon|x}(e|x)$$

$$\text{Since } m(d, x) = \mathbb{E}(Y | d, x) = \mathbb{E}(\varphi(d, x, \varepsilon) | d, x) = \int \varphi(d, x, e) \cdot f_{\varepsilon|D, X}(e|d, x) de$$

$$\text{then we have. } = \int \varphi(d, x, e) f_{\varepsilon|x}(e|x) de$$

$$\text{Then, we differentiate w.r.t. } d \text{ as } m'_d(d, x) = \int \varphi'_d(d, x, e) f_{\varepsilon|x}(e|x) de$$

$$\text{Given that } \theta_{CATE}(d, x) = \mathbb{E}(\varphi'_d(d, x, \varepsilon) | d, x)$$

$$= \int \varphi'_d(d, x, e) \cdot f_{\varepsilon|D, X}(e|d, x) de.$$

$$\text{Since } f_{\varepsilon|x}(e|x) = f_{\varepsilon|D, X}(e|d, x) \text{ given that } D, \varepsilon \text{ independent}$$

$$\text{then we have } \theta_{CATE}(d, x) = m'_d(d, x)$$

... Because $D \perp \varepsilon | X$, at a particular level of X , the remaining unobservables, ε ,

distributed the same across values of D . then, moving d , the only thing that changes in $\mathbb{E}(Y | d, x)$ is the structural outcome function, not selection into D .

... then, the slope of $m(d, x)$ w.r.t. d is the average structural marginal effect $\theta_{CATE}(d, x)$

Using law of iterated expectations, we have that

$$\theta_{ATE} = \mathbb{E}(\varphi'_d(D, X, \varepsilon)) = \mathbb{E}(\underbrace{\mathbb{E}(\varphi'_d(D, X, \varepsilon) | D, X)}_{\theta_{CATE}(D, X)})$$

for continuous treatments... $m(d, x)$ does encode causal info

Hence, $\theta_{ATE} = \mathbb{E}(\theta_{CATE}(D, X)) = \mathbb{E}(m'_d(D, X))$... averaging the marginal effect over the empirical distribution of (D, X) gives overall ATE ... through its slope in d .

2. The strength shall be $(2) \rightarrow (1) \rightarrow (3)$.

$(2) \rightarrow (1)$: if ε is independent from X and $\mathbb{E}(\varepsilon) = 0$

then conditioning on X does not change the distribution of ε , hence $\mathbb{E}(\varepsilon|X) = \mathbb{E}(\varepsilon) = 0$

$(1) \rightarrow (2)$: a counter-example can be provided: $X \in \{0, 1\}$, $P(X=1) = P(X=0) = \frac{1}{2}$

and $U \sim N(0, 1)$ independent of X , $\varepsilon := (1+X) \cdot U$

we then have that $\mathbb{E}(\varepsilon|X) = (1+X) \cdot \mathbb{E}(U|X) = (1+X) \cdot 0 = 0 \dots (1)$ holds

However, ε is not independent given that U and X are independent

of X , as the spread of ε changes with X . $\varepsilon|X=0 \sim N(0, 1)$, but $\varepsilon|X=1 \sim N(0, 4)$

(for instance, $P(|\varepsilon| > 2 | X=0) \neq P(|\varepsilon| > 2 | X=1)$). hence (2) does not hold.

$(1) \rightarrow (3)$: using law of iterated expectations,

$\mathbb{E}(\varepsilon X) = \mathbb{E}(\mathbb{E}(\varepsilon X | X)) = \mathbb{E}(X \cdot \mathbb{E}(\varepsilon | X))$, as we have $\mathbb{E}(\varepsilon | X) = 0$,

$\mathbb{E}(\varepsilon X) = \mathbb{E}(X \cdot 0) = 0$. Furthermore, $\mathbb{E}(\varepsilon) = \mathbb{E}(\mathbb{E}(\varepsilon | X)) = \mathbb{E}(0) = 0$, hence (3) holds.

$(3) \rightarrow (1)$: an counter-example can be provided: $X \sim N(0, 1)$, and $\varepsilon := X^2 - \mathbb{E}(X^2) = X^2 - 1$

Then, we have $\mathbb{E}(\varepsilon) = 0$, provided that $\mathbb{E}(X^2) = 1$. Furthermore, $\mathbb{E}(\varepsilon X) = \mathbb{E}(X^3 - X)$

$= \mathbb{E}(X^3) - \mathbb{E}(X) = 0 - 0 = 0 \dots (3)$ holds. However, $\mathbb{E}(\varepsilon | X) = X^2 - 1 \neq 0$, $\forall X$, hence (1) fails

Since we know $\mathbb{E}(\varepsilon | X) = 0$ in $Y = \beta_0 + \beta_1 X + \varepsilon$, as we can have uncorrelatedness

from mean-independence, we know $\mathbb{E}(\varepsilon X) = 0$ and $\mathbb{E}(\varepsilon) = 0$.

Hence, $\mathbb{E}(Y) = \beta_0 + \beta_1 \mathbb{E}(X) + \mathbb{E}(\varepsilon) = \beta_0 + \beta_1 \mathbb{E}(X)$

Then, provided that $\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X) \cdot \mathbb{E}(Y)$

$$= \mathbb{E}(\beta_0 X + \beta_1 X^2 + X\varepsilon) - \mathbb{E}(X) \cdot (\beta_0 + \beta_1 \mathbb{E}(X)) = \beta_0 \mathbb{E}(X) + \beta_1 \mathbb{E}(X^2) + \underbrace{\mathbb{E}(X\varepsilon)}_{=0 \text{ from mean indep.}} - \beta_0 \mathbb{E}(X) - \beta_1 (\mathbb{E}(X))^2 = \beta_1 (\mathbb{E}(X^2) - (\mathbb{E}(X))^2) = \beta_1 \text{Var}(X)$$

Hence, $\beta_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$, Hence $\beta_0 = \mathbb{E}(Y) - \beta_1 \mathbb{E}(X) = \mathbb{E}(Y) - \frac{\text{Cov}(X, Y)}{\text{Var}(X)} \cdot \mathbb{E}(X)$

3. Knowing the full regression must satisfy the orthogonality of X and ϵ

∴ $X^T \epsilon = 0$. Hence, $X^T (y - X_1 b_1 - X_2 b_2) = 0$

$$\begin{pmatrix} X_1^T \\ X_2^T \end{pmatrix} (y - X_1 b_1 - X_2 b_2) = \begin{pmatrix} X_1^T y - X_1^T X_1 b_1 - X_1^T X_2 b_2 \\ X_2^T y - X_2^T X_1 b_1 - X_2^T X_2 b_2 \end{pmatrix} = 0$$

Therefore, $\begin{cases} X_1^T y = X_1^T X_1 b_1 + X_1^T X_2 b_2 & \dots i \\ X_2^T y = X_2^T X_1 b_1 + X_2^T X_2 b_2 & \dots ii \end{cases}$

Since $P_1 = X_1 (X_1^T X_1)^{-1} X_1^T$, and from i, we have $X_1^T X_1 b_1 = X_1^T y - X_1^T X_2 b_2$

$$b_1 = \underbrace{(X_1^T X_1)^{-1}}_{X_1^+ P_1} X_1^T y - \underbrace{(X_1^T X_1)^{-1} X_1^T X_2}_{X_1^+ P_1} b_2 \dots X_1 b_1 = P_1 y - P_1 X_2 b_2$$

∴ Substituting into ii, then $X_2^T X_2 b_2 = X_2^T y - X_2^T X_1 b_1 = X_2^T y - X_2^T (P_1 y - P_1 X_2 b_2)$

∴ $\underbrace{X_2^T (I - P_1)}_{M_1} X_2 b_2 = \underbrace{X_2^T (I - P_1)}_{M_1} y$. Since $M_1 = I - P_1$, there is $X_2^T M_1 X_2 b_2 = X_2^T M_1 y$

Provided that P_1, M_1 are projection matrices. $M_1^T = M_1, M_1^2 = M_1$

Hence, $X_2^T M_1 X_2 b_2 = X_2^T M_1^T M_1 X_2 b_2 = (\underbrace{M_1 X_2}_{\tilde{X}_2})^T M_1 X_2 b_2 = \tilde{X}_2^T \tilde{X}_2 b_2$

$$X_2^T M_1 y = (\underbrace{X_2^T M_1^T}_{\tilde{X}_2^T}) (\underbrace{M_1 y}_{\tilde{y}})$$

$\tilde{X}_2$ by definition

$$\left. \begin{aligned} \dots \tilde{X}_2^T \tilde{X}_2 b_2 &= \tilde{X}_2^T \tilde{y} \\ b_2 &= (\tilde{X}_2^T \tilde{X}_2)^{-1} \tilde{X}_2^T \tilde{y} \end{aligned} \right\}$$

∴ Frisch-Waugh Theorem: partial out X_1 from both y and X_2 , then regress residuals on residuals can attain coefficient estimated b_2 for X_2 .

From the full regression, residual $e = y - X_1 b_1 - X_2 b_2$

give that $M_1 X_1 = (I - P_1) X_1 = X_1 - X_1 (X_1^T X_1)^{-1} X_1^T X_1 = X_1 - X_1 = 0$

$$\underbrace{M_1 y}_{\tilde{y}} = M_1 X_1 b_1 + M_1 X_2 b_2 + M_1 e = \underbrace{M_1 X_2}_{\tilde{X}_2} b_2 + M_1 e \dots \tilde{y} = \tilde{X}_2 b_2 + M_1 e$$

give that $X^T e = 0, X_1^T e = 0$ as a result. Then, as $P_1 e = X_1 (X_1^T X_1)^{-1} \underbrace{(X_1^T e)}_{=0} = 0$

$M_1 e = (I - P_1) e = e - 0 = e$. Therefore, $\tilde{y} = \tilde{X}_2 b_2 + e$

∴ residuals from $\tilde{y}$ on $\tilde{X}_2$ equal to the full residuals e .

(P_1 projects e onto $C(X_1)$, because of the orthogonality of X_1 and ϵ , the projection must be zero)

if, instead, we pursue $\hat{b}_2 = (\tilde{X}_2^T \tilde{X}_2)^{-1} \tilde{X}_2^T y$, which only residualize the regressor but not the dependent variable, and y still contains X_1 -component,

$$1st\ y = P_1 y + M_1 y = P_1 y + \hat{y}. \text{ Then, we have } \tilde{X}_2^T y = \tilde{X}_2^T (P_1 y + \hat{y}) = \tilde{X}_2^T P_1 y + \tilde{X}_2^T \hat{y}$$

$$\text{Since } \tilde{X}_2 = M_1 X_2, \tilde{X}_2^T P_1 y = (M_1 X_2)^T P_1 y = X_2^T M_1^T P_1 y = X_2^T \underbrace{M_1 P_1}_{\text{as } M_1 \text{ is a projection matrix}} y = 0$$

($M_1 = I - P_1$, hence $M_1 P_1 = P_1 - P_1^2 = P_1 - P_1 = 0$, as P_1 is a projection matrix, too)

$$\text{Hence, } \tilde{X}_2^T y = \tilde{X}_2^T \hat{y}, \hat{b}_2 = (\tilde{X}_2^T \tilde{X}_2)^{-1} \tilde{X}_2^T y = b_2 \dots \text{ per part c.}$$

... as $\tilde{X}_2$ is orthogonal to the entire space spanned by X_1 , it "can't see" $P_1 y$.

as $P_1 y$ lies in the column space of X_1 , its projection onto $\tilde{X}_2$ is zero. ... slope unchanged.

Considering residual, we have from $\hat{b}_2$, there is $\hat{e} = y - \tilde{X}_2 \hat{b}_2 = y - \tilde{X}_2 b_2$

$$= (P_1 y + \hat{y}) - \tilde{X}_2 b_2 = P_1 y + e, \text{ unless } P_1 y = 0, \text{ we have } \hat{e} \neq e. \dots \text{ No}$$

as $\hat{y} = \tilde{X}_2 b_2 + e$

$$\text{Then, for SSR, as } \widehat{SSR} = \|\hat{e}\|^2 = (e + P_1 y)^T (e + P_1 y)$$

$$= e^T e + 2e^T P_1 y + (P_1 y)^T (P_1 y) = e^T e + (P_1 y)^T (P_1 y) = \|e\|^2 + \|P_1 y\|^2 = SSR + \|P_1 y\|^2 \geq SSR$$

as OLS residual satisfies $X_1^T e = 0$, hence $e^T P_1 = 0$, $e^T P_1 y = 0$

with equality only if $P_1 y = 0$, y orthogonal to X_1 ... No

$$\text{Provided that } \hat{y} = \tilde{X}_2 b_2 + e, \text{ then } \hat{y}^T \hat{y} = (\tilde{X}_2 b_2 + e)^T (\tilde{X}_2 b_2 + e)$$

$$= (\tilde{X}_2 b_2)^T (\tilde{X}_2 b_2) + e^T e + 2(\tilde{X}_2 b_2)^T e = b_2^T \tilde{X}_2^T \tilde{X}_2 b_2 + e^T e, \hat{y}^T \hat{y} - e^T e = b_2^T \tilde{X}_2^T \tilde{X}_2 b_2$$

$$\text{give that } \tilde{X}_2 = M_1 X_2, X_2^T e = 0, (\tilde{X}_2 b_2)^T e = 0$$

$$\text{Since we know that } b_2 = (\tilde{X}_2^T \tilde{X}_2)^{-1} \tilde{X}_2^T \hat{y}, b_2^T \tilde{X}_2^T \tilde{X}_2 b_2 = \hat{y}^T \tilde{X}_2 [(\tilde{X}_2^T \tilde{X}_2)^{-1}]^T (\tilde{X}_2^T \tilde{X}_2)$$

$$\begin{aligned} \cdot (\tilde{X}_2^T \tilde{X}_2)^{-1} \tilde{X}_2^T \hat{y} &= \hat{y}^T \tilde{X}_2 (\tilde{X}_2^T \tilde{X}_2)^{-1} \tilde{X}_2^T \hat{y} = \hat{y}^T M_1 X_2 (X_2^T M_1 X_2)^{-1} X_2^T M_1 \hat{y} = \hat{y}^T X_2 (X_2^T M_1 X_2)^{-1} X_2^T \hat{y} \\ &= (M_1 X_2)^T M_1 X_2 = X_2^T \underbrace{M_1^T M_1}_{M_1^2 = M_1, \dots \text{proj. matrix}} X_2 = X_2^T M_1 X_2 \\ &= M_1 \cdot (M_1 y) = M_1 y = \hat{y} \end{aligned}$$

Because $\hat{y} = M_1 y$ is orthogonal to the column space of X_1 , then $X_1^T \hat{y} = X_1^T M_1 \hat{y} = (M_1 X_1)^T \hat{y} = 0$. With zero coefficient and fitted values, residuals are $\hat{y}$. Hence $SSR_1 = \|\hat{y}\|^2 = \hat{y}^T \hat{y}$

By the previous part, we have that $\hat{y} = \underbrace{\hat{X}_2}_{b_2} b_2 + e$. Hence the residuals are e . Hence $SSR_2 = \|e\|^2 = e^T e$. $b_2 = (\hat{X}_2^T \hat{X}_2)^{-1} \hat{X}_2^T \hat{y}$

Applying Frisch-Waugh Theorem, regression 3 can be expressed as

$\hat{y} = X_1 \hat{b}_1 + X_2 \hat{b}_2 + \hat{e}$, as $\hat{y} = M_1 y$, we have $M_1 X_1 = 0$, $X_1^T \hat{y} = 0$.

... X_1 has no explanatory power for $\hat{y}$. Then, we have that $\hat{b}_1 = 0$.

$\hat{y} = X_2 \hat{b}_2 + \hat{e}$ same as the previous part, $\hat{b}_2 = b_2$, $\hat{e} = e$. Hence

$$SSR_3 = \|\hat{e}\|^2 = \|e\|^2 = e^T e.$$

For regress 4, a numerical counter-example can be provided:

$$X_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad X_2 = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} \quad y = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \quad P_1 = X_1 (X_1^T X_1)^{-1} X_1^T = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad M_1 = \frac{1}{3} \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$$

$$\hat{y} = M_1 y = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \quad \hat{X}_2 = M_1 X_2 = \frac{1}{3} \begin{pmatrix} -2 \\ -2 \\ 4 \end{pmatrix}, \quad \text{then we can calculate } SSR_2 = \frac{1}{2},$$

and $SSR_4 = 1$ which is a numerical counter-example that $SSR_4 \neq e^T e$.